



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

Name: JOHN ROBERT P. ELCA	Date: 9/24/26
Section: BSIT 2A	Instructor: MAXYNE NUELA IGNACIO

- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

The program is designed to manage IT automation incident tickets using a Python list. It allows the user to add, display, search, remove, and count active incident tickets. And anytime the user can exit and end the program. The program uses simple list operations and sequential traversal to organize and manage the tickets.

- II. **Program Operations:** Briefly explain how your program performs each operation.
(Insert Specific Code Line and Your Explanation)

Add Incident:

```
tickets.append([incident_id, bot, description])
```

The append() function adds a new incident ticket to the end of the tickets list and can be shown in the next functions.

Display Incidents:

```
for ticket in tickets:
```

```
    print(ticket[0], ticket[1], ticket[2])
```

The for loop goes through each ticket in the list and displays its Incident ID, Bot, and Description in a linear way so that it's easy to see.

Search Incident:

```
for ticket in tickets:
```

```
    if ticket[0] == incident_id:
```

```
        print("Incident Found:", ticket)
```

It checks each ticket one by one and compares its Incident ID with the ID entered by the user, that's what the program do. If they match, the incident details are displayed.

Remove Resolved Incident:

```
tickets.remove(ticket)
```

When the matching Incident ID is found, remove() deletes that ticket from the list because it has been resolved.

Count Active Incidents:

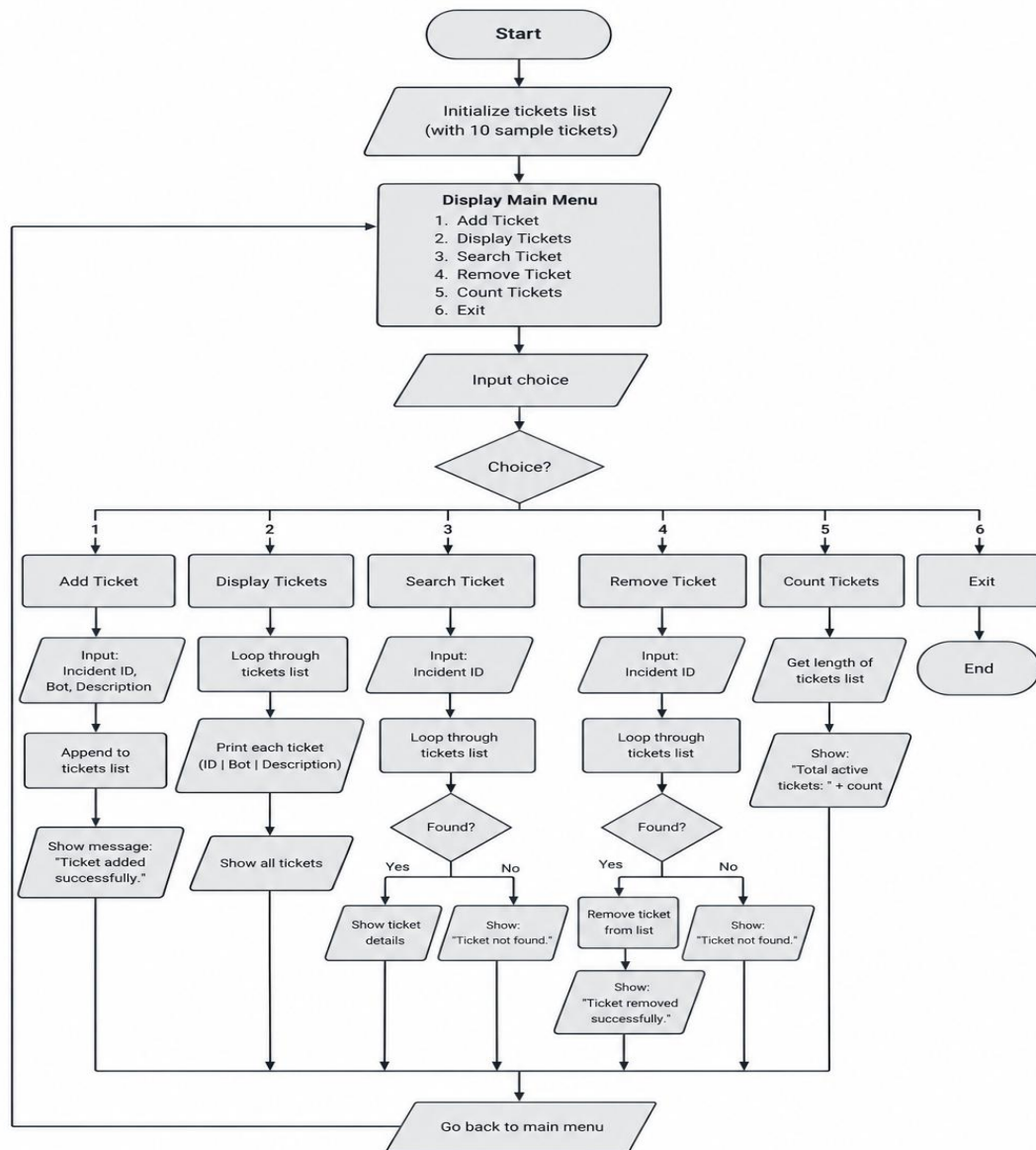
```
len(tickets)
```

This len() function counts the number of tickets currently stored in the list, giving the total number of active incidents or items that are in the list.



College of Computer Studies

III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.



IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

This activity helped me understand how a linear data structure can be used in a real IT-related program. It has a huge difference when using it, you can see clearly the data and it's even more useful when you're working with larger projects. The Python list stores the incident tickets in a sequential order, allowing the program to add, display, search, remove, and count tickets. I also learned that simple list operations and loops can be used to manage incident records effectively and efficiently
